Project Report Template: Markov Model Based Project

Cover Page

Include: university name and logo, course title and code, project title, student name(s) and ID(s), section, course teacher, submission date.

Abstract

150–250 words: the problem, the AI method used, the main design, and the key result in numbers.

1. Introduction

1.1 **Background and motivation:** *why this problem matters, preferably in a Bangladesh context.*

1.2 **Problem statement:** *the exact problem in 2–3 sentences.*

1.3 **Objectives:** *3–4 clear, measurable objectives.*

1.4 **Scope and limitations:** *what the project covers and what it does not.*

2. Literature Review

Review 4–6 related works and summarise them in a table. End with the gap your project addresses.

Author (Year)	Method	Data / Problem	Result	Limitation

3. Problem Description

Describe the real-world system, its users and its inputs and outputs. Explain what process changes over time or in sequence, and why a probabilistic, state-based model fits.

4. Tools and Environment

Programming language, libraries (e.g. scikit-fuzzy, NumPy, hmmlearn, Pygame), IDE and hardware.

5. Model Formulation

- *Markov chain or Hidden Markov Model (HMM)? Use a chain if the states are observed directly; use an HMM if the states are hidden and only observations are seen.*
- *States (and observations for an HMM), with their meaning.*
- *Markov assumption: why the next state depends only on the current state, and whether this holds.*
- *State transition diagram, with arrows labelled by probabilities.*

6. Parameter Estimation

- *Initial probabilities (π), transition matrix (A), and emission matrix (B) for an HMM.*
- *How they were obtained: counted from data, or from experts or literature.*
- *Data source and how sequences were prepared (cleaning, discretising continuous values).*
- *Check that each row of every matrix sums to 1.*

7. Algorithms and Worked Example

- *Markov chain: n-step prediction (A^n), sequence probability, steady-state distribution ($\pi A = \pi$).*
- *HMM: Forward algorithm, Viterbi algorithm, and Baum–Welch if parameters are learned.*
- *Solve **one example by hand**: a 2–3 step prediction and a steady-state calculation, or a Forward/Viterbi trellis for 3 observations.*

8. Implementation, Results and Analysis

- *Code snippets and screenshots.*
- *Predicted vs actual sequences on test data; accuracy, log-likelihood, or a confusion matrix for HMM decoding.*
- *Steady-state distribution and its real-world meaning (e.g. "in the long run, 60 % of days are sunny").*
- *Comparison with a baseline (e.g. always predicting the most frequent state).*
- *How well the Markov assumption holds, and how predictions change when the probabilities change.*

9. Discussion

Interpret the results: why the system behaved as it did, its strengths and weaknesses, and any failed or unexpected cases. Mention ethical or social issues if relevant.

10. Conclusion and Future Work

Summarise what was achieved against each objective, then give 2–3 concrete improvements.

11. References

At least 5 references in IEEE style.

12. Appendix

Full source code or GitHub link, extra results and a short user manual.

Contribution Table (Group Projects Only)

Student ID	Name	Contribution